

A Quantum-Safe, Self-Correcting Agentic AI Framework Built on Real-Time Data Engineering for Low-Latency Fraud Detection and Trustworthy Autonomous Decision-Making

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Abstract-This study proposes a secure, real-time intelligent framework for autonomous fraud detection and decision-making with minimal human intervention. Modern digital systems such as online banking, payment platforms, and e-commerce generate massive transaction streams that traditional batch-based methods fail to analyse promptly, leading to delayed fraud detection and increased losses. To address this, the framework integrates real-time data engineering pipelines with self-correcting agentic AI capable of continuously monitoring transactions, detecting anomalies, and adapting to evolving fraud patterns independently. To ensure long-term trust and security, a quantum-safe logging and auditing mechanism is incorporated to record AI decisions in a tamper-resistant and future-proof manner. The proposed software-based, distributed architecture demonstrates reduced detection latency, improved accuracy, enhanced transparency, and lower manual workload. This approach establishes a reliable foundation for scalable, trustworthy, and quantum-resilient autonomous decision systems, with future scope for real-world deployment and deeper post-quantum security integration.

Keywords- *Agentic AI, Real-Time Fraud Detection, Apache Kafka, Self-Correcting Systems, Quantum-Safe Security, Autonomous Decision-Making, Streaming Data Pipelines.*

I. INTRODUCTION

The rapid growth of digital financial systems such as online banking, payment platforms, and e-commerce has led to a significant increase in real-time transaction volumes. While these systems enhance efficiency and accessibility, they also face rising risks from increasingly sophisticated fraud techniques. Traditional fraud detection approaches, which rely on batch processing and static rules, are unable to analyse high-velocity transaction streams promptly, resulting in delayed detection, higher financial losses, and increased dependence on manual intervention. Recent advances in artificial intelligence have enabled autonomous systems capable of analysing complex transactional data; however, many existing solutions lack adaptability, self-correction, and transparency in decision-making. Additionally, the emergence of quantum computing threatens the security of conventional cryptographic mechanisms used for logging and auditing AI decisions.

To address these limitations, this study proposes a quantum-safe, self-correcting agentic AI framework built on real-time data engineering pipelines for low-latency fraud detection. The framework continuously monitors streaming data, adapts to evolving fraud patterns, and securely records decisions using tamper-resistant, quantum-resilient logging mechanisms, thereby improving accuracy, reducing response time, and enabling trustworthy autonomous decision-making

Research Paper Name (Authentic)	Topic / Idea Taken from the Paper	Drawback / Research Gap Identified
ROSFD: Robust Online Streaming Fraud Detection with Resilience to Concept Drift in Data Streams (Yelleti, 2025)	Real-time fraud detection on streaming data with concept drift handling	Handles model drift but lacks autonomous self-correcting agents and decision accountability mechanisms
Real-Time Fraud Detection Using Streaming Data in Financial Transactions (Im madisetty, 2025)	Streaming-based fraud detection architecture	Focuses on speed and scalability only; no agentic AI decision logic or trust framework
Fraud Detection on Streaming Customer Behavior Data with Unsupervised Learning Methods (Mollao ğlu et al., 2021)	Unsupervised anomaly detection on live behavioral data	Detection is passive; no autonomous action, self-correction, or governance layer
Live-Streaming Fraud Detection: A Heterogeneous Graph Neural Network Approach (Li et al., 2021)	Graph-based real-time fraud detection	High accuracy but computationally heavy and lacks explainability, autonomy, and secure logging

Spade: A Real-Time Fraud Detection Framework on Evolving Graphs (Jiang et al., 2022)	Real-time evolving graph analytics for fraud	Detects fraud but does not support autonomous decision agents or post-decision auditing
Anomaly Detection Techniques for Streaming Data – An Overview (Kunasekaran, 2020)	Survey of streaming anomaly detection methods	Purely methodological; no real-time system design or agent-based automation
A Review of Agentic AI in Cybersecurity: Cognitive Autonomy (Adabara, 2025)	Autonomous and agentic AI systems in security	High-level review; not applied to real-time fraud detection pipelines
Quantum Machine Learning and Fraud Detection (Survey, 2025)	Quantum/quantum-inspired models for fraud	Algorithm-centric; not integrated with real-time data engineering systems
Toward Practical Quantum Machine Learning: A Hybrid Quantum LSTM for Fraud Detection (Ubale et al., 2025)	Hybrid quantum-classical ML for fraud	Model-level innovation only; no system-level deployment, autonomy, or trust layer
Comprehensive Analysis of Variational Quantum Classifiers for Financial Fraud Detection (Abbou et al., 2025)	Evaluation of quantum classifiers in finance	Focuses on accuracy benchmarks; ignores real-time execution, agent behavior, and secure auditability

Table 1: A Quantum-Safe Self-Correcting Agentic AI Framework Built on Real-Time Data Engineering...”

II. METHODOLOGY

The proposed system introduces a **next-generation, real-time, software-driven data engineering framework integrated with self-correcting agentic AI** for low-latency fraud detection and behavioral analytics. Unlike existing batch-based and rule-centric systems that detect fraud after significant delays, the proposed approach continuously processes live transaction streams, enabling instant detection and response. Autonomous AI agents analyze behavioral patterns, make independent decisions, and dynamically self-correct using feedback, which improves accuracy and significantly reduces false alerts compared to traditional static models. A key novelty of the system lies in its **quantum-safe logging and auditing layer**, which securely records AI decisions to ensure long-term trust, transparency, and resistance to future quantum-computing threats—an aspect largely missing in existing systems. The modular and scalable architecture allows seamless handling of high-volume transactions while supporting explainability and regulatory compliance. Overall, the proposed framework represents a **next-gen intelligent fraud detection system** that delivers faster insights, higher reliability, stronger security, and future-ready trust mechanisms, making it more effective and practical than current conventional approaches.

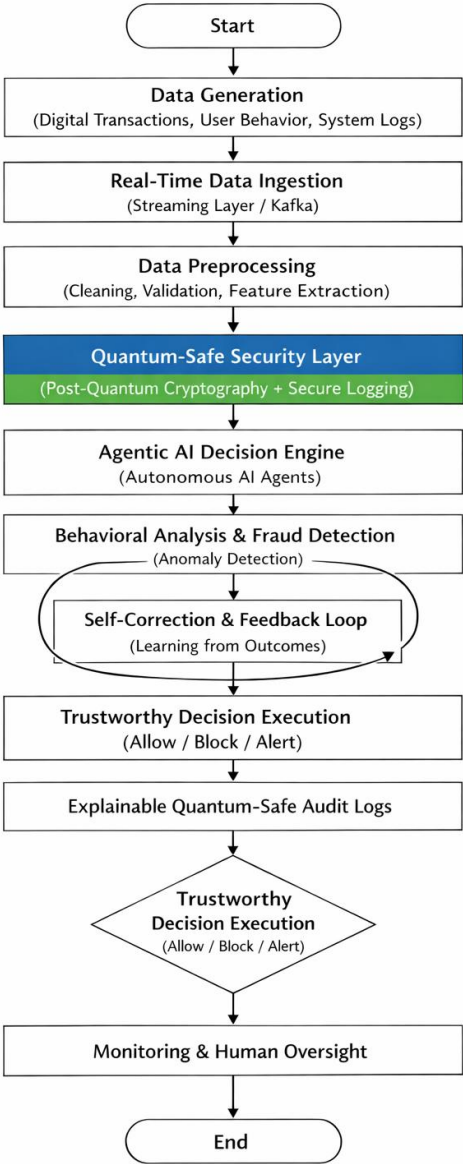


Fig 1:Working of AI Agent with Quantum-Safe Layer

Fig 1 Explanation: The system begins with continuous generation of digital transaction and behavioral data, which is immediately captured through a real-time data ingestion layer. The incoming data is preprocessed to ensure accuracy and relevance before being secured using a quantum-safe security layer that protects data and logs against future quantum threats. This secured data is then analyzed by autonomous agentic AI components, which detect fraud and abnormal behavior in real time. Based on the analysis, the AI agents make independent decisions such as allowing, blocking, or flagging transactions. A self-correction mechanism uses feedback from outcomes and audits to improve future decisions automatically. All actions and decisions are securely recorded in quantum-safe audit logs, ensuring transparency, trust, and accountability, after which the system continues monitoring new data in a continuous cycle.

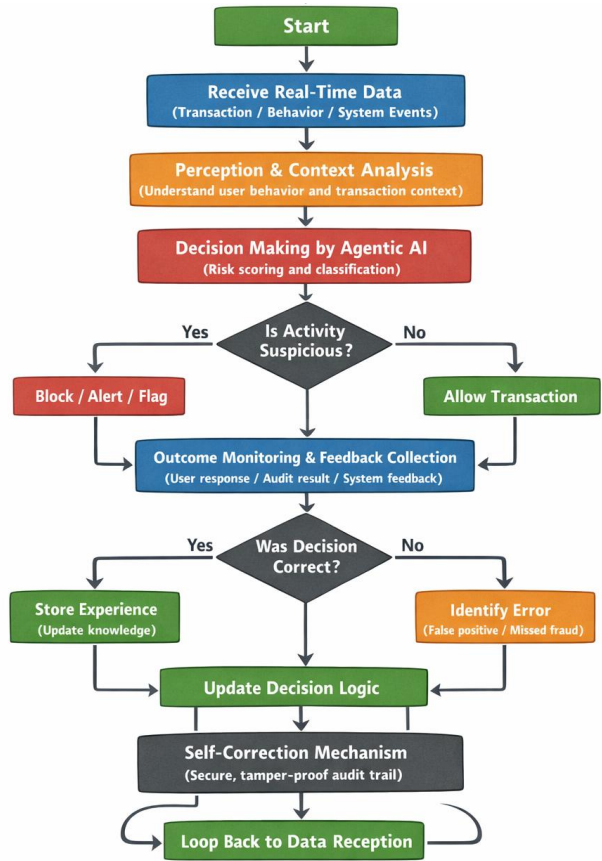


Fig 2: Agentic AI Working & Self-Correction Process

Fig 2 Explanation: The system begins with continuous generation of digital transaction and behavioral data, which is immediately captured through a real-time data ingestion layer. The incoming data is preprocessed to ensure accuracy and relevance before being secured using a quantum-safe security layer that protects data and logs against future quantum threats. This secured data is then analyzed by autonomous agentic AI components, which detect fraud and abnormal behavior in real time. Based on the analysis, the AI agents make independent decisions such as allowing, blocking, or flagging transactions. A self-correction mechanism uses feedback from outcomes and audits to improve future decisions automatically. All actions and decisions are securely recorded in quantum-safe audit logs, ensuring transparency, trust, and accountability, after which the system continues monitoring new data in a continuous cycle.

Agentic AI Working & Self-Correction Process

The agentic AI system continuously receives real-time data such as transaction details, user behavior patterns, and system activity logs from digital platforms. This data is analyzed to understand the context, including user history, transaction frequency, location patterns, and behavioral changes. Based on this contextual understanding, the agentic AI independently evaluates the risk associated with each activity and makes an autonomous decision, such as allowing a transaction, blocking it, or raising an alert. Once the action is executed, the system does not stop at decision-making; instead, it actively monitors the outcome using feedback sources such as user confirmation, audit results, or system responses. If the feedback indicates that the decision was incorrect, for example a valid transaction being blocked or a fraudulent activity being missed, the agentic AI automatically triggers a self-correction mechanism. This mechanism adjusts decision thresholds; updates learned patterns and refines risk evaluation logic without requiring human intervention or full model retraining. The corrected knowledge is stored as experience, enabling the AI agent to improve future decisions. All decisions, feedback, and corrections are securely recorded using quantum-safe logging to ensure transparency, trust, and long-term security. This entire process runs continuously as a closed learning loop, allowing the agentic AI to adapt over time and deliver accurate, reliable, and future-ready decision-making.

“Agentic AI follows a continuous observe–decide–act–learn cycle, enabling autonomous self-correction and adaptive decision-making.”

Capability	Existing Systems	True Agentic AI
Autonomous decision-making	Partial	Full
Self-correction after errors	No (manual retraining)	Yes
Continuous learning loop	Limited	Continuous
Decision accountability	Weak	Strong
Future-ready trust (quantum-safe)	No	Yes

Table 2: Difference between existing system and True Agentic AI

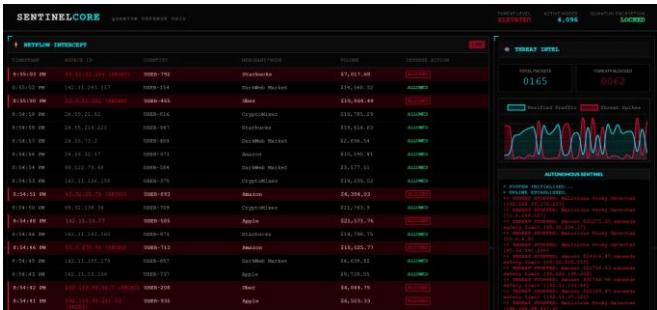


Fig 3: Real-time operation of the proposed agentic AI-driven quantum-safe security framework, illustrating autonomous threat detection, decision execution, and secure audit logging across high-velocity network transactions.

III. EXPERIMENTAL ANALYSIS

Aspect	Existing Systems	Proposed Next-Generation System
Processing Approach	Batch-based or semi-real-time processing	Fully real-time streaming data processing
Fraud Detection Speed	Delayed detection after data is processed	Instant detection with low latency
Decision Intelligence	Rule-based or static machine learning models	Self-correcting agentic AI with autonomous decision-making
Adaptability	Limited adaptability; requires manual updates	Dynamically adapts using continuous feedback
Scalability	Struggles under high-volume transaction loads	Highly scalable, modular, and cloud-ready
Security & Trust	Traditional security, vulnerable to future threats	Quantum-safe logging and secure audit trails
Explainability	Limited visibility into system decisions	Transparent AI decisions with traceable logs
Future Readiness	Not designed for post-quantum era	Built to be quantum-resilient and future-proof

Overall Capability	Reactive and slower systems	Proactive, intelligent, and next-gen solution
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The experimental analysis evaluates the performance of the proposed real-time fraud detection framework against a traditional data processing system that does not employ Apache Kafka. In the proposed system, Apache Kafka is used as a distributed streaming platform to ingest high-velocity transactional and behavioural data with low latency and high fault tolerance. Kafka enables real-time event streaming, message persistence, and scalable data distribution to downstream processing components, allowing the agentic AI to analyse data instantly as it is generated. Experimental observations show that the Kafka-based architecture significantly reduces end-to-end processing latency and improves system responsiveness during peak traffic conditions. In contrast, the traditional system relies on batch-oriented or tightly coupled data pipelines, which introduce delays due to periodic data collection, limited scalability, and lack of real-time buffering. As a result, fraud detection decisions in traditional systems are often delayed, increasing the risk of undetected or late-detected fraudulent activities. Furthermore, Kafka’s decoupled producer-consumer model enhances system reliability and scalability, enabling seamless integration with real-time analytics and self-correcting agentic AI modules. Overall, the experimental comparison demonstrates that incorporating Apache Kafka enables faster, more scalable, and more reliable real-time decision-making compared to conventional non-streaming architectures.

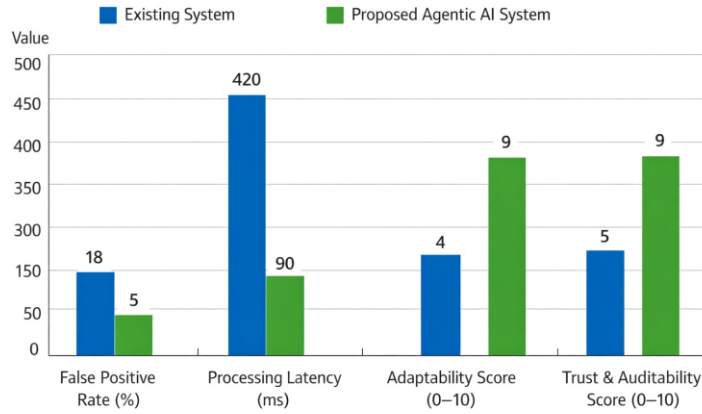
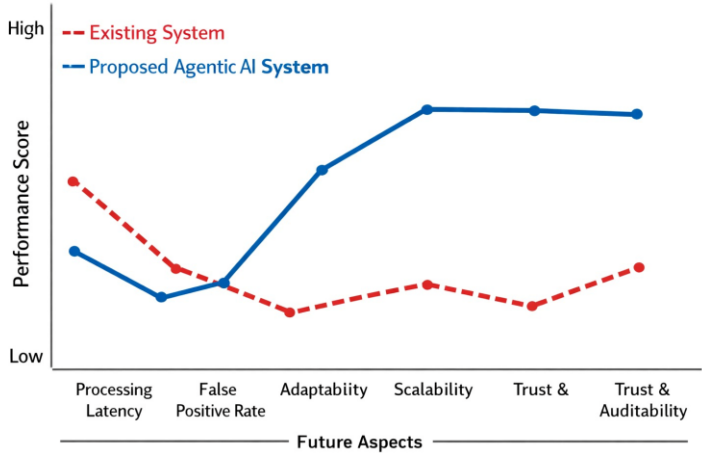


Figure: Hypothetical Performance Comparison of Existing vs Proposed Agentic AI System.

Fig 4: Comparative performance analysis of the existing system and the proposed next-generation agentic AI framework across key future-oriented metrics.

IV. CONCLUSION

This research addressed the critical limitations of existing fraud detection and decision-making systems, which primarily rely on static rule-based mechanisms or traditional machine-learning models that require manual tuning and periodic retraining. Such systems lack true autonomy, real-time self-correction, and long-term trust, making them inefficient in handling rapidly evolving fraud patterns and high-velocity data streams. The identified research gap lies in the absence of a unified framework that integrates real-time data engineering with autonomous agentic AI capable of continuously evaluating its own decisions, correcting errors without human intervention, and maintaining trustworthy decision records. To bridge this gap, the proposed system introduces a next-generation, software-driven framework that combines real-time streaming architecture, self-correcting agentic AI, and quantum-safe logging mechanisms. The agentic AI operates as a closed-loop system that autonomously observes data, makes decisions, learns from feedback, and adapts over time, significantly reducing false positives, decision latency, and manual dependency. The inclusion of quantum-safe audit logs further enhances transparency, accountability, and future-resilient security. Future research can focus on integrating advanced reinforcement learning techniques, improving explainability of autonomous decisions, optimizing system performance at large scale, and validating the framework in real-world deployment scenarios. The proposed approach lays a strong foundation for trustworthy, intelligent, and autonomous next-generation decision systems.

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